Meta Quest

Meta Motion Accessories

TEAM 8

Shristi Keshri

Herman Yuan

Hanlin Hu

Ismail Guled

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1. Business and Enterprise Analysis

Meta Platforms, Inc. famously known as Meta and previously branded as The Facebook Inc. is an American multinational social technological enterprise that is based in Menlo Park, California. It was founded on January 4, 2004, in Cambridge, Massachusetts, US by Mark Zuckerberg, Eduardo Saverin, Andrew McCollum, Dustin Moskovitz, and Chris Hughes. Meta is very widely known for offering various services, some of which are Facebook, Instagram, Messenger, Threads, Whatsapp, Meta Quests, Horizon Worlds, etc.

1.1 Research of Enterprise

1.1.1 Industry Segment

Meta's primary focus is developing applications and technologies that have been helping people connect with loved ones, find communities, and grow various types and scales of businesses. Meta Platforms are very involved in advertisements, augmented reality (AR), and virtual reality (VR).

Diving deeper into AR and VR, Meta has acquired Oculus which is a part of Reality Labs. Meta Quest which was called Oculus Quest until 2022. It is a segmentation of the company that is a line of virtual reality headsets with augmented reality features. The first model was developed and released on May 21, 2019, and since then it has been updated and released multiple times Meta also has the new Horizon OS. This was Meta's effort to develop immersive AR and VR technologies which seem to be gaining popularity in gaming, entertainment, education, and professional training industries.

1.1.2 Products and Services

Meta Quest is primarily known for its cutting-edge virtual reality hardware and software products and these include these products and services:

- **Meta Quest or Oculus Quest Headsets** are standalone VR devices designed for immersive gaming, professional training modules, fitness and social entertainment utilities. The current lineup includes Meta Quest 2 and Meta Quest 3.
- **Horizon Worlds** is another service that Meta provides which is a virtual reality space where friends can form communities and worlds and hang out and interact with other users.

- **Meta Workplace** is one of the other services Meta Inc. provides to its users and is mainly VR tools that focus on business improvement and collaboration, training modules, and virtual meetings.

The service proposed by our team to enhance, introduces **Meta Motion Accessories** which plans on including devices targeting the lower body - ankles, knees and waist. This aims to enhance physical activities and fitness. It would enrich the gaming experience of users by enabling natural movement that would be tracked more efficiently allowing users to immerse themselves in the virtual environment wholly.

1.1.3 Market

The Meta Quest influences and impacts many markets and arenas of our daily lives.

- **Gaming and Entertainment:** The consumer market is influenced by the VR gaming people, people who are casual gamers or professional ones. These people have the main focus on entertainment and interaction with communities through immersive techniques.
- **Professional:** The people who work and are business professionals use the VR of Meta Workplace for virtual meetings, professional training, remote collaboration, pair programming, and more.
- **Educational:** Meta has widespread usage in educational institutes as schools, and universities use VR for interactive and highly evolved learning, study tours, simulations, and laboratory experiments.
- **Fitness:** With the addition of Meta Motion Accessories, users can engage in full body workouts and fitness challenges with increased accuracy as all of their motions would be tracked.
- **HealthCare and Rehabilitation:** VR applications for improving health and providing therapy and fitness has seen to increase and with our tools it would grow rapidly as it tracks and encourages accurate body movement.

1.1.4 Business and Organizational Structure

The company's market has a global reach and is very prominent in countries like Europe, Asia, and the United States, which are adopting the services this technology has to offer in both entertainment and professional usage.

Being such a big company with over ten services, the services are divided into divisions. Meta Quest and Horizon World fall under the Reality Labs division. It is a sect of the company that focuses on the research and development of AR and VR tech that includes wearable headsets and spatial multidimensional systems. The organizational structure of this company is a hybrid of functional and product-based divisions. They include:

- **CEO, COO, CTO, CPO, and the enterprise officials** who oversee all the services the company provides including the Reality Labs.
- **Reality Labs division** focuses mainly on the development and improvement of augmented and virtual realities. They are also responsible for the marketing and overseeing of the AR-VR tech which mainly includes Meta Quest.
- The sole focus of the **product team** is the technology they are developing so the Reality Labs would typically have three - one for Meta Quest, one for Horizon Worlds and a new one for Meta Motion Accessories.
- **Specification teams** would be in each product team which would individually handle the hardware, software, AI, ML, spatial computing, mechanics, and more according to the developers' specializations and areas of expertise.
- **Publicity and sales team** which would help advertise and market the Meta Motion Accessories to bring in more investments and usage by consumers.

1.1.5 Value Chain

The Value chain of Meta Quest is as follows:

- **Research and Development:** A lot of money is invested in virtual and augmented reality.
- **Manufacturing:** Meta Quest has to have a lot of partner companies that aid them in the manufacturing of the hardware for the headsets and more.
- **Marketing and Advertising:** The service needs to be publicized to get more people to buy it and management of sales is also needed.
- **Distribution:** The product needs to be sold and needs to be set off to different partner stores or online platforms like Amazon, Meta Quest Store, etc.
- **Customer Service:**
- Help the existing customers with troubleshooting issues and repairs and updates.

1.1.6 Stated Areas of Differentiation

Meta Quest offers unique features like affordability, standalone headsets, educational and professional usability and versatility and claims that set the enterprise apart from all the other ones in the market. The areas of differentiation that set Meta Motion Accessories apart and make it preferable to be used over the existing products are as follows:

- **Full-Body Integration**: Enhancing movement tracking for fitness and gaming as now it will track upper and lower body movements making it very accurate.
- **Enhanced Fitness Features**: Supporting interactive workouts and real-world activity monitoring which would encourage users to achieve their goals in the right way.
- **Immersive Gaming**: Elevating the gaming experience with natural movements that sync seamlessly with virtual environments.

All these features add a competitive advantage to the Meta Motion Accessories and are projected to increase the usage of people by a lot.

1.2 Five Forces Analysis

1.2.1 Bargaining Power of Buyers

In this case, buyers are people who purchase VR devices primarily for entertainment.

- As customers have many alternatives (Apple, Microsoft, Google, HTC, HP, Sony, etc.) in the market, they do have bargaining power with buyers. *Impact on the bargaining power of buyers: increases*
- Consumers are highly sensitive to price and product quality. Consumers may switch to other competitors if Meta increases prices or fails to deliver compelling new experiences. *Impact on the bargaining power of buyers: increases*
- At present, VR devices are more focused on providing an experience rather than practical utility, which may lead consumers to opt for more practical devices. *Impact on the bargaining power of buyers: increases*

    Bargaining power of buyers: high

Overall: Within the entertainment segment, the bargaining power of those who purchase VR devices primarily for entertainment is high. They do have many choices, either in the market or out of the market. However, with Meta Motion Accessories, the value proposition increases as those features are unavailable in other existing products.

1.2.2 Bargaining Power of Suppliers

In this case, suppliers are people who provide hardware manufacturing and computing platforms.

- The number of suppliers relative to VR firms is low. Meta relies on specialized suppliers for its hardware components, such as lenses, sensors, processors, mics, etc. Suppliers like Qualcomm have the power of suppliers due to the limited number of firms producing high-end chips. *Impact on the bargaining power of suppliers: increases*
- Cost of hardware customization. Unlike traditional products, Meta needs to set requirements for suppliers, asking them to customize or re-develop hardware for their VR devices. Suppliers like Qualcomm need to develop a brand new chip that can work on the Quest platform while its major chip platform is Android. This will largely increase the costs of suppliers. *Impact on the bargaining power of buyers: increases*
- Limited alternative suppliers. There are very few suppliers capable of producing hardware for high-end VR devices. It is difficult to find a chip supplier in the market that can replace Qualcomm. *Impact on the bargaining power of buyers: increases*
- Meta Quest 3 accounts for over 70% of the market. The large scale gives it some leverage in negotiating prices. *Impact on the bargaining power of buyers: decrease*
- The cost of the computing platform is low. A self-developed computing platform saves a significant amount in both R&D and maintenance costs. *Impact on the bargaining power of buyers: decrease*

  Bargaining power of suppliers: moderate-to-high

Overall: The overall bargaining power of suppliers is moderate-to-high. Within the hardware manufacturing segment, the bargaining power of suppliers is high as Meta highly relies on hardware suppliers. Within the computing platform, the bargaining power of suppliers is low. By expanding its accessory line, Meta could be able to decrease the bargaining power of its suppliers by increasing production volume of in-house equipment.

1.2.3 Threat of New Entrants

- The technology barrier to entry into this market is high. Major technology companies (Apple, Microsoft, Google, HTC, HP, Sony, etc.) have major resources to develop and complete their VR products. However, new entrants will struggle without mature technology.
- The investment threat for new entrants is high due to capital investment in technology development, software ecosystems, hardware manufacture, distribution networks, advertisement, etc. Competitive companies (Apple, Microsoft, Google, HTC, HP, Sony, etc.) have already established market positions, with access to resources, content, market, and mature technology.

- Due to the saturation of the current market and the lack of significant differentiation among products, the introduction of new devices from new entrants is unlikely to generate a substantial market response.

  Threat level: low-to-moderate

Overall: There is a low-to-moderate threat of new entrants. The existing strength of Meta and its competitors (Apple, Microsoft, Google, HTC, HP, Sony, etc.) makes the market difficult to break into. The high technological barriers, significant investment needs, and market saturation make it challenging for new entrants to compete with established companies in the VR market.

1.2.4 Threat of Substitute Products

- Other entertainment methods. Since most consumers currently purchase VR devices primarily for entertainment, they also have alternative options such as traditional gaming consoles, mobile games, or emerging augmented reality (AR) devices, which may appeal to a similar audience. Additionally, traditional 2D media and social experiences serve as substitutes for the immersive experience offered by VR.
- Platforms of use make consumers buy different devices. Since most consumers of VR devices are regular consumers (eg., not for commercial, government, education, healthcare, and enterprise), there are many similar high-end products in the market such as Apple Vision Pro, Sony PlayStation VR, Valve Index VR, HTC Vive Pro, etc as Meta has not integrated all the platforms of use. Consumers could choose other devices based on their favorite platforms of use. For instance, those focused on the PlayStation platform would choose Sony, those focused on the Steam platform would opt for Valve, and those wanting to experience the latest VR and AR-integrated technology would go for Apple.
- High cost-effectiveness for Meta Quest 3. Although there are many similar products in the market, Meta Quest 3 is recognized as the most cost-effective high-end VR device in the market. Its affordability ($429.99) combined with powerful performance has allowed this device to maintain over half of the high-end VR devices market share. (For other VR deceptive sets: PlayStation VR 2: $550. Apple Vision Pro: $3,499. HP Reverb G2: $469. Valve Index: $999)

  Threat level: moderate-to-high

Overall: The threat of substitute products in the VR devices market is moderate-to-high. Since Meta has not integrated all VR platforms, consumers tend to consider their actual usage scenarios when purchasing devices. Meta's current advantage lies in its high cost-effectiveness, attracting many consumers through low pricing and high performance.

1.2.5 Competitive Rivalry within the Industry

- Meta faces significant competitors in the industry such as Apple, Microsoft, Google, HTC, HP, Sony, etc. The rivalry is intense with competitors' product releases, price competition, and enhanced experience.
- Meta maintains high cost-effectiveness in the industry so it differentiates itself through its affordability. It tries to attract customers based on its low price and high performance.
- Technology development is the major competition in the market. Better technology used in VR devices could largely enhance customers' experience. Based on Facebook and cutting-edge R&D, Meta has led technology development and research.

  Threat level: moderate-to-high

Overall: The threat of competitive rivalry within the industry is moderate-to-high. Although Meta Quest has the highest cost-effectiveness in the industry, it faces many strong competitors which may have better R&D in the future.

1.3 Evaluation

Meta's stated strategy is justified by its competitive position. There are five dominant factors for the analysis of a VR device.

- **Technology:** Meta focuses on building its technological lead through research and development (R&D), acquisitions of high-tech companies, collaboration with colleges and research institutions, open-source engagement, etc. Technology development will largely enhance customers' experience.
- **Pricing:** Meta focuses on the affordability and cost-effectiveness of their products, which helps it dominate the consumer VR market.
- **Ecosystem:** Meta's development in creating its VR ecosystem with a lot of social, entertainment, and fitness applications, has made the ecosystem functional and rich.
- **Performance:** Meta Quest's overall performance is strong in the market.
- **Design:** Customers are sensitive to VR devices' weight and comfortability. Meta Quest is known for its lightweight and comfortability.

The analysis of Porter's Five Forces demonstrates that Meta's high bargaining power over suppliers and buyers validates its business strategy so far. The stated strategy is justified by its competitive position. The threat of new entrants into the industry is low but the threat of substitute products is high. Meta's stated strategy is justified by its competitive position as all companies like Meta will face such a situation. At last, the threat of competitive rivalry within the industry is moderate-to-high as technology development varies rapidly.

1.4 Software Application Portfolio

1.4.1 New Feature Proposals

- **Full-Body Gaming Integration**
  By enabling a more immersive gaming experience, Meta's lower-body devices, like motion trackers for the ankles, knees, and waist, have the potential to completely transform gaming. Games become more involved and physically engaging with the help of these peripherals, which allow for natural motions like sprinting, jumping, and squatting. To completely combine pleasure and fitness, picture an adventure game where players must physically leap over obstacles or duck to avoid them. In addition to improving the user experience, this innovation promotes exercise. Meta can further its goal of creating immersive, movement-driven virtual worlds while drawing in gamers looking for active play by fusing health and gaming.
- **Motion-Integrated Fitness Games**
  Meta Motion Accessories, which provide guided workouts with real-time feedback, can improve fitness applications. Using lower-body accessories, users can track specific movements, such as squats or lunges, ensuring proper form and maximizing effectiveness. Users are encouraged to stay active by fitness games that leverage these gadgets to make working out fun and gamified. Leaderboards and prizes, together with challenges and objectives catered to individual fitness levels, encourage participation and consistency.
- **Augmented Reality Fitness/Biometrics**
  Meta Motion Accessories' integration of real-time biometrics gives users a comprehensive fitness experience. These gadgets provide comprehensive workout insights by measuring things like heart rate, steps, and calories burned. When used in conjunction with an AI coach, users get goal-setting and tailored feedback to maximize performance. Exercise becomes an engaging experience when augmented reality and fitness are combined, with users being guided by virtual obstacles or landscapes while they walk around the real world. This creative strategy combines technology and fitness, establishing Meta as a leader in developing products that promote exercise and give users useful health insights.

1.4.2 Portfolio Justification

These features capitalize on Meta's existing VR infrastructure while targeting new user segments. Fitness and gaming enthusiasts represent growing markets, and *Meta Motion Accessories* create unique opportunities for engagement without requiring significant hardware changes. It also achieves the goal of innovation that Meta has alongside achieving the goal of low cost due to the ease of implementation of these potential features

1.4.3 Application to Develop Further

Motion-Integrated Fitness Games ought to be given priority since they take advantage of virtual reality's immersive qualities and capitalize on the expanding fitness industry. Adoption among a variety of audiences can be encouraged by partnerships with fitness influencers and gamified exercise challenges.

1.5 Five-Forces Analysis Evaluation

Bargaining Power of Buyers:

- The analysis identifies that buyers have high bargaining power due to multiple alternatives in the market (Apple, Microsoft, Google, etc.), price sensitivity, and the ability to find similar products for both entertainment and professional uses. The Meta Quest's competitive pricing and the ability to offer immersive entertainment or practical utilities help Meta retain customers, justifying their strategy of focusing on affordability and a wide range of applications. However, some customers might need the utility part of the product to be more practical such as business collaboration tools, leading to possible loss in the buyer group.

Bargaining Power of Suppliers:

- The analysis acknowledges the product has a high reliance on specialized hardware suppliers, such as Qualcomm, which makes Meta dependent on limited suppliers, increasing supplier power. However, the self-developed computing platform by Meta can balance out the cost brought by limited suppliers with a reduced overall cost for hardware. This justifies Meta's strategic investments in proprietary technologies and R&D to reduce the reliance on suppliers.

Threat of New Entrants:

- The analysis identifies high technological and financial barriers for new entrants. As a result, the immediate risk from new entrants will be reduced. As most of the products in the area are similar, Meta's lead in AR and VR technology through strategic acquisitions is key to maintaining its competitive edge.

Threat of Substitute Products:

- The analysis identifies the moderate-to-high threat of substitute products, including traditional gaming consoles, mobile games, and augmented reality (AR) devices like the Apple Vision Pro and PlayStation VR. These alternatives offer similar entertainment experiences and attract consumers away from VR. However, Meta's primary advantages are the cost-effectiveness and performance of the Meta Quest 3. The product is significantly cheaper than many high-end substitutes. This competitive pricing along with strong performance, let Meta effectively mitigate the risk of consumers switching to substitutes, ensuring its continued appeal in the market.

Competitive Rivalry:

- The analysis identifies strong competitive rivalry from major players such as Apple, Microsoft, and Sony, especially in terms of R&D and technological advancements. However, Meta differentiates itself through its focus on cost-effective, high-performance products, allowing it to compete strongly with these industry giants. Meta's strategies are well-aligned to counter the rivalry by maintaining a competitive price and continuing cutting-edge technology investments.

1.6 Traceability to the Application Portfolio

The *Five Forces Analysis* aligns with the proposed software application portfolio, as follows:

- **Affordability and Cost-Effectiveness**: Meta keeps its products affordable while adding new functionalities, such as **Augmented Reality Fitness/Biometrics** or **VR integration into social media**, which aligns with the company's goal of maintaining cost leadership in the market. These applications and integrations add relatively very low cost to the product line, as its strategy increases the value of the product without a significant rise in costs.
- **Ecosystem and Performance**: The integration of VR into social media fits perfectly into Meta's ecosystem of social platforms like Facebook and Instagram. By expanding Quest's software ecosystem, Meta enhances its

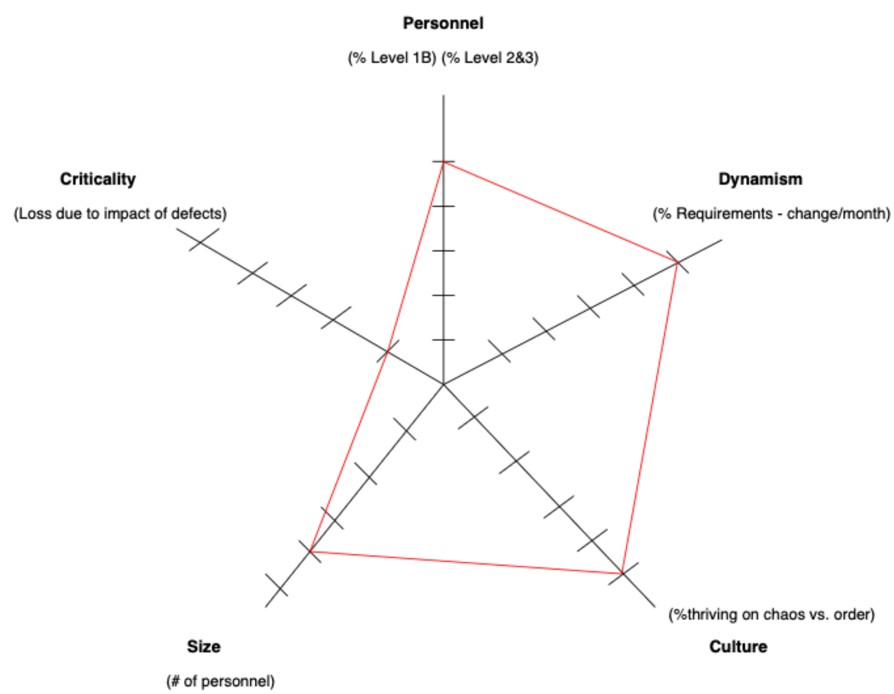
ecosystem's functionality and gains a huge advantage in its VR product from those who lack such social media integration.

- **Technological Lead**: The Android APP Support Expansion makes the Meta Quest ecosystem more versatile and allows Mata to further strengthen its competitive position in the market. Meta Quest can remain the leading platform by supporting additional apps and appealing to broader customers without requiring substantial hardware changes.

Thus, by introducing *Meta Motion Accessories*, Meta evolves beyond entertainment into fitness and holistic VR experiences, solidifying its leadership in the market.

2. Intended Software Engineering Process

2.1 Spider Diagram



Spider Diagram for our Intended Software Engineering Project

Personnel

The Meta Oculus VR will have the need to hire specialized hardware engineers so that they can design ergonomic and comfortable designs. Thus, as this requires skilled personnel, this would rate around high.

Criticality

When dealing with the health of an individual, it is always a critical situation and in order to make products to improve it we need to be very accurate and thus, in order to succeed, this would be highly critical. It improves the lifestyle of people and can be said that our product focuses on comfort.

Dynamism

The market that focuses on health oriented VR is hardly available and thus the features would need to change rapidly in order to accommodate the evolving needs of the clients and thus dynamism would be rated high.

Size

When compared to meta, a smaller segment is dedicated to Quest and when dealing with the new accessories, it would be even smaller and hence, we could rate this at a medium.

Culture

Meta has a culture of innovation and has a commitment to its users to provide the best experience and with its introduction of health-oriented plans and accessories, it would also accommodate a greater crowd thus, this would also be likely high.

2.2 Project Management

Release Plan

A release plan will be developed to define the main milestones and features for the motion-tracking accessory. This plan will provide a roadmap for the project timeline, specifying when major functionalities, such as core tracking capabilities and gesture customization, will be delivered. It will help prioritize features and allocate resources effectively to meet the project's objectives.

Iteration Plan

An iteration plan will be created for each release cycle. This plan will break down the current release's functional and non-functional requirements into manageable tasks. The team will decide on weekly priorities and execute these tasks incrementally, ensuring progress aligns with the overall release plan and project goals.

Risk Plan

A risk plan will be prepared at the start of the project. It will outline potential risks, their likelihood, and their impact. The plan will include strategies to mitigate these risks, as well as contingency actions to address them if they occur. Examples of risks include hardware delays, integration issues, and budget overruns. This proactive approach will help minimize disruptions and ensure a smooth project timeline..

Project Schedule
A project schedule will be created to align with the release plan. It will outline key activities and milestones for each phase of the project, including inception, elaboration, construction, and transition. The schedule will be updated as the project evolves, ensuring that the team remains on track to meet delivery dates for each release.

2.3 Requirements

Storyboards
Storyboards will illustrate how users interact with the motion-tracking accessory. They will provide a clear narrative to stakeholders and guide the development of features by visualizing user scenarios.

Problem Statement
The problem statement will define the key issues faced by Meta Quest users, such as motion-tracking inaccuracies and usability challenges. It will ensure a shared understanding of the project goals among all team members.

Business Case
The business case will evaluate the financial viability of the motion-tracking accessory by estimating potential revenue increases, expanded market reach, and competitive differentiation against project costs.

Use Cases
Use cases will depict how various user groups (gamers, designers, educators) interact with the system. A use case diagram will provide a visual overview of these interactions to ensure alignment with user needs.

User Stories and Scenarios
The requirements will be broken into user stories, describing the desired functionality from the perspective of end users, developers, and administrators. These stories will guide the system's design and implementation.

Non-Functional Requirements
Non-functional requirements will specify essential characteristics such as low latency,

lightweight design, and compatibility with existing Meta Quest platforms. These requirements will ensure the system supports the functional features effectively.

Acceptance Plan
The acceptance plan will define the criteria for project completion. It will include performance benchmarks, user satisfaction metrics, and integration success with the Meta Quest ecosystem. Also, some associated system tests will validate that the final product meets the acceptance criteria. These tests will include motion accuracy, data transmission reliability, and overall user experience, ensuring a robust and reliable system.

2.4 Analysis

2.4.1 Description

Analysis involves understanding the problem and exploring solutions for the motion-tracking accessory. Work products such as problem analysis, solution analysis, and diagrams will refine our solution and ensure alignment with project objectives.

2.4.2 Problem Analysis

A structured problem analysis document will be produced to define the scope, constraints, and objectives. This includes identifying key user needs, system requirements, and areas of improvement that inform subsequent design and development tasks.

2.4.3 Solution Analysis

A solution analysis document will outline potential approaches that align with identified requirements. It will compare and evaluate solution strategies, establish selection criteria, and support decision-making for the final design direction. The sequence diagram and state diagram will be used to describe the interactions between the product and the Meta Quest system.

2.5 Architecture

2.5.1 Subsystem Model

The system will be divided into several key subsystems to meet functional and non-functional requirements. And sequence diagrams will illustrate how these components interact to achieve system goals.

2.5.2 Target Environment

The solution will function within the Meta Quest ecosystem and integrate with third-party applications. It will support development and deployment in Unity/Unreal Engine and operate across platforms compatible with Meta Quest.

2.5.3 System Architecture

Subsystems are designed to ensure modularity and scalability. This architecture will enable a responsive and immersive user experience while maintaining compatibility with existing systems.

3. Requirements Analysis

3.1 Problem Statement

Meta Quest 3 cannot directly track users' precise actions and foot movements as it primarily relies on the headset's built-in cameras and the tracking system of the controllers to capture head and hand movements. However, the Meta Quest 3's controllers have removed tracking rings and adopted a gesture tracking mechanism, which significantly improves gesture control accuracy compared to the Meta Quest 2. Users still experience tracking loss in fast-motion scenarios when they try to hit three or four buttons quickly. Consequently, users who play VR games must carry one or two controllers (104 grams each) when using the device. At the same time, they have to monitor the controllers' battery levels constantly.

3.2 Problem Analysis

There are 3 primary types of actors in this situation. The first type is the individual users or teams who use Meta Quest 3 mainly for VR games Those users need enhanced motion feedback and traction when playing VR games. They may experience a loss of tracking accuracy, loss of tracking stability, and failure to detect their movements when playing VR games. Although the Meta Quest 3 only tracks head and hand movements, users may lack motion feedback while Sony PlayStation VR and KAT VR Walkers provide.

The second type is the schools or companies that use Meta Quest 3 for training and education. Motion tracking inaccuracies may cause students or trainees to feel frustrated when making mistakes during VR operations or training. In VR scientific experiments or VR training, inaccurate tracking can result in incorrect execution of experimental steps or simulated training that does not align with the correct real situations. Thus it will affect this type of actors' understanding and learning outcomes.

The final type is the designers and artists. They mainly use the Meta Quest 3 for software development and art creation. When they work on VR sculpting, VR painting, or VR modeling, inaccurate tracking of gestures and tools can result in imprecise control over lines, shapes, colors, or details. This will affect the quality and efficiency of their art creations.

3.3 Business Case

Costs

- Labor Cost: Labor is the primary cost of developing an enhanced motion-tracking accessory for the Meta Quest. Assuming a development team of 20 engineers, each with an average yearly salary of $150,000, the rounded total labor cost for this team per year would be approximately $3,000,000.

- Research & Development Cost: The R&D phase would cost additional resources as it is a brand-new feature used for Meta Quest. Costs for new materials, hardware testing, software adaptation, and iterations could be approximately $5,000,000.

- Manufacturing and Distribution Cost: There could be costs associated with manufacturing, production, packaging, storage, and distribution. Assuming the initial production is 100,000 units, the per-unit production cost is about $75, and the total cost is about $750,000.

- Maintenance and Support Cost: The new accessor may confuse customers. Maintenance and customer support would be necessary after launch. The annual maintenance cost could be around $1,000,000.

- Training Cost: Training salesmen or engineers is necessary as well. The estimated cost could be around $1,000,000.

Benefits

- Increased Meta Quest Sales: An enhanced motion-tracking accessory could significantly enhance the sales of Meta Quest. Assuming this accessory could drive additional sales by 5% of the estimated 20 million existing and potential users around the world, this would translate to 1,000,000 additional units sold, which is about $499,999,000 excluding tax.

- Accessory Sales: Assuming this accessory is sold at $99 each and 10% of the estimated 20 million existing users are willing to buy this accessory. This accessory could drive an additional $198,000,000 excluded tax.

3.4 Solution Analysis

This new motion-tracking accessory is designed to fix the biggest issues of Meta Quest 3 players by enhancing motion capture for accurate footsteps and general movement. The accessory will be a direct companion to the existing Meta Quest 3 platform to deliver a superior user experience in multiple applications such as games, training, and even design.

This solution analysis is about system-to-system interactions and it is user-centric and doesn't go in deep to the internal system details. It's also a thorough examination of the communications between components in the system and agents outside the system to compensate for the motion-tracking constraints.

3.4.1 Actors and Their Interactions

1. **Individual Users/Teams (Gamers)**
    - Users who engage with the Meta Quest 3 headset for gaming require high-precision tracking of hand and foot movements for an immersive experience.
    - Interaction: Gamers will wear the accessory, which will transmit real-time motion data to the Meta Quest 3 headset. The system will process this data to deliver improved control and provide haptic feedback for enhanced immersion.
2. **Schools/Companies (Students/Trainees)**
    - Educational or training institutions use Meta Quest 3 for VR-based training sessions, requiring precise movement capture to reduce errors and frustration.
    - Interaction: The accessory will facilitate better movement tracking during training, allowing students to execute tasks more accurately and receive real-time visual and tactile feedback during simulations.
3. **Designers/Artists**

- Designers and artists utilize the Meta Quest 3 for creative endeavors like VR sculpting, painting, and modeling. They need consistent and stable gesture recognition.
- Interaction: Artists will use the accessory while working on VR creations. The accessory will ensure their hand and foot gestures are accurately tracked, providing a consistent environment for working with fine details.

4. **Meta Quest 3 System**
    - Meta Quest 3 is the central VR system that will integrate input from the enhanced motion-tracking accessory to optimize user experiences.
    - Interaction: The system interfaces with the accessory to receive precise motion data, process it, and then translate the data into an immersive VR experience for the user.
5. **Enhanced Motion-Tracking Accessory**
    - The accessory enhances the headset's ability to track body movements beyond head and hand movements, particularly focusing on foot positioning and gestures.
    - Interaction: The accessory will continuously track body movement and relay data to the Meta Quest 3 system. It also has the ability to provide haptic feedback to users.

3.4.2 Interface-Level Interactions

The solution operates through a combination of wireless connectivity, real-time data transmission, and feedback mechanisms. Below are the primary interactions at the system interface level

Accessory to Meta Quest 3 System

- Motion Data Transmission: The accessory uses Bluetooth to communicate with the headset, transmitting motion data including precise positions, speed, and direction of movements.
- Data Integration: The Meta Quest 3 system integrates this motion data with its VR rendering, ensuring user movements translate accurately in the virtual environment.

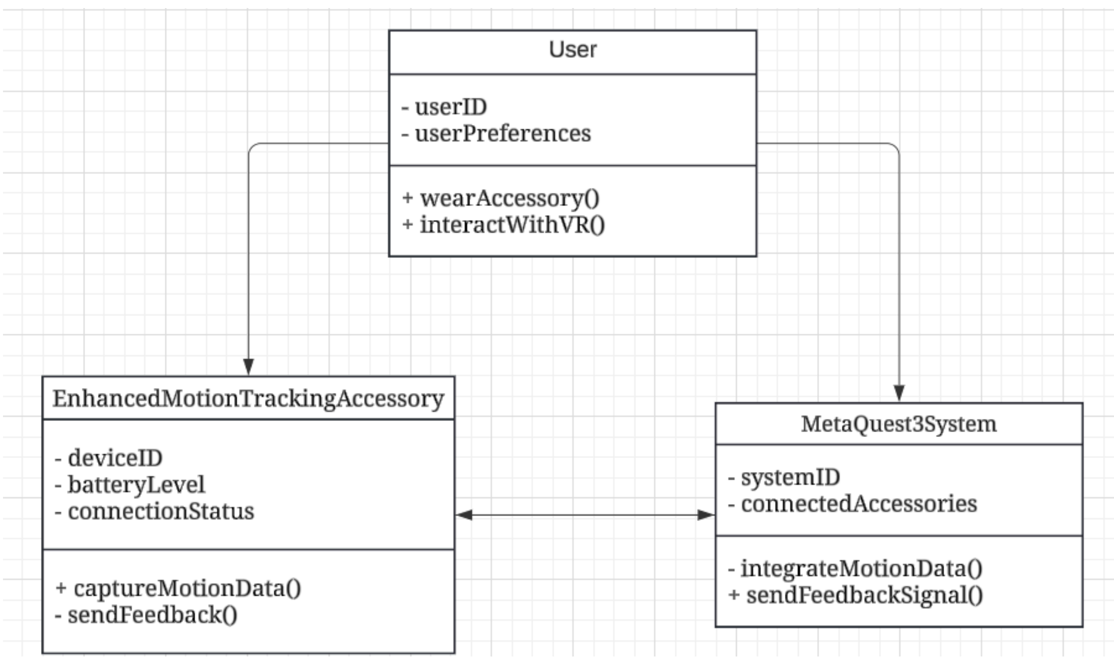
Meta Quest 3 System to Accessory

- Feedback Signals: The system sends feedback signals (e.g., vibration instructions) to the accessory to provide haptic feedback when specific actions or events occur in the virtual space.
- Accessory Status Updates: The system also provides notifications about the accessory's status, such as battery levels and connection quality, to keep users informed.

User to Accessory

- User Input: Users wear or attach the accessory (e.g., leg bands or foot sensors) to enable additional motion capture capabilities.
- Haptic Feedback: The accessory delivers tactile responses to users during
- gameplay or training, increasing the feeling of immersion.

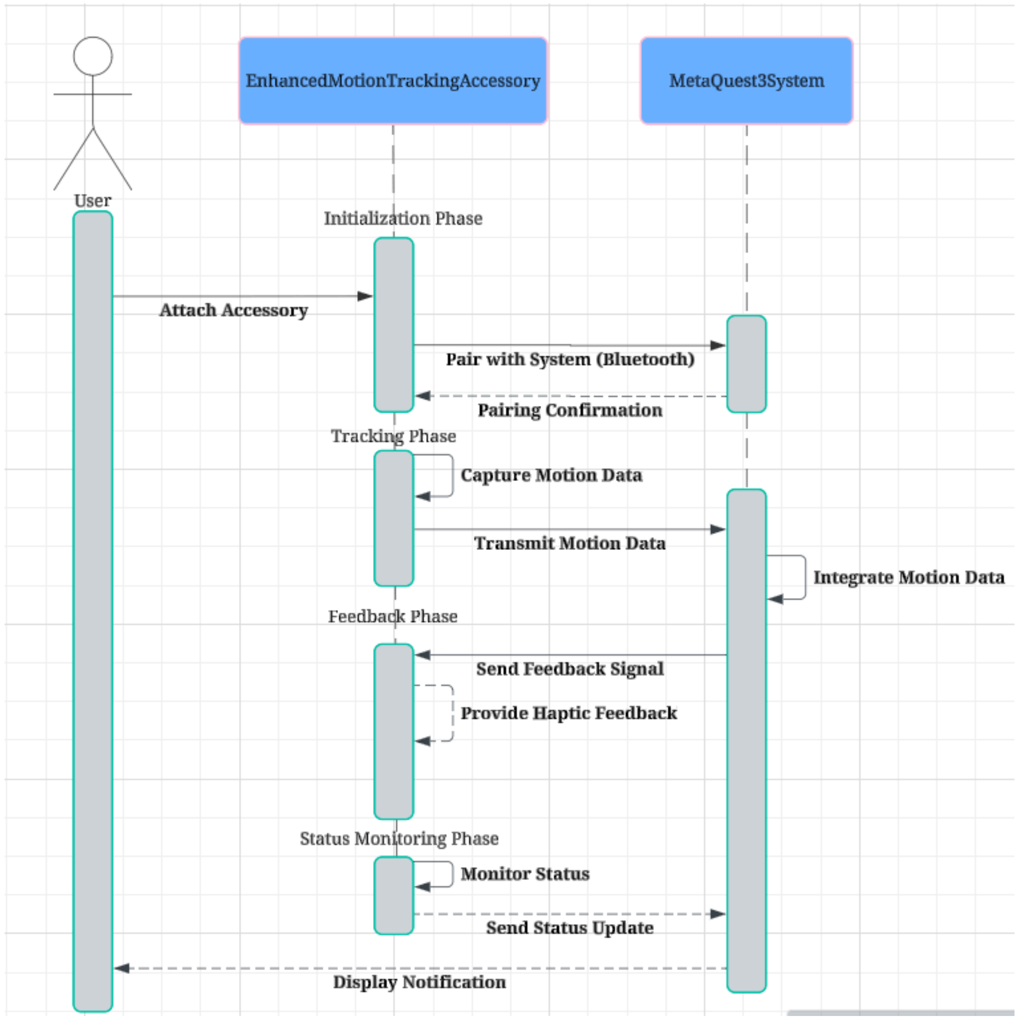
3.4.3 Class Model for Interface-Level Interactions



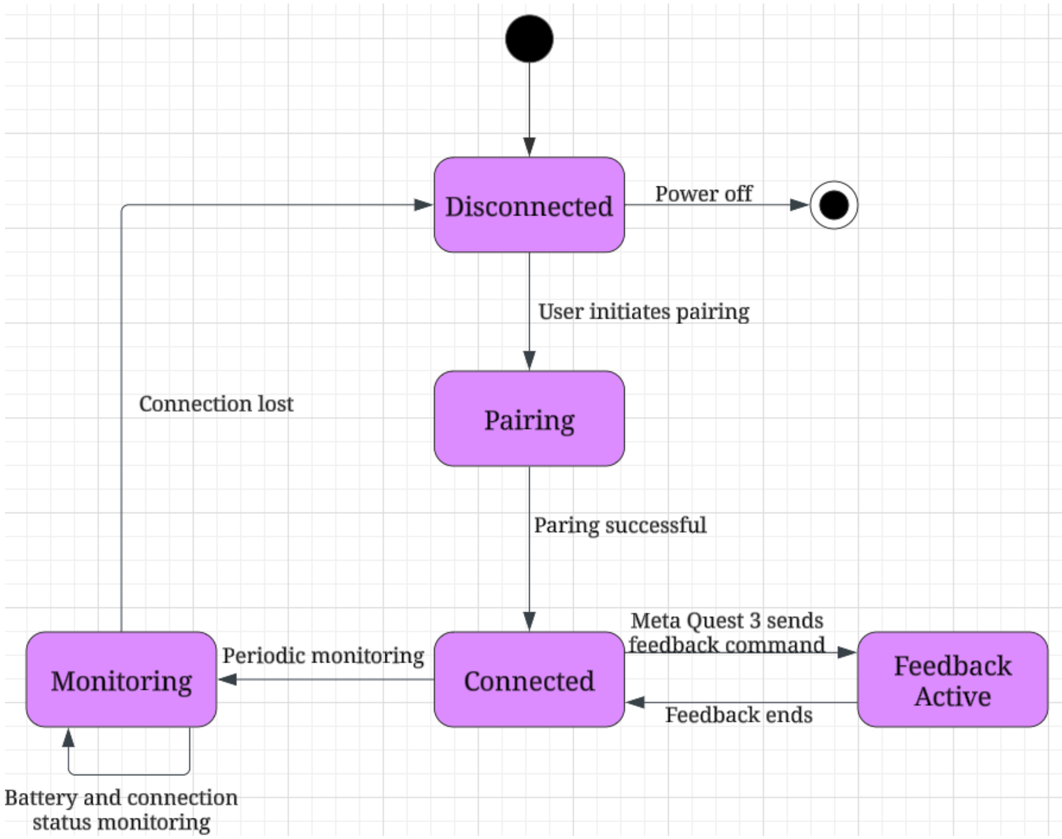
1. **EnhancedMotionTrackingAccessory**
    - **Attributes**:
        - deviceID: Identifier for the accessory.
        - batteryLevel: Current battery level.
        - connectionStatus: Bluetooth connection status.
    - **Methods**:
        - captureMotionData(): Captures real-time motion data and prepares it for transmission.

        - sendFeedback(): Activates haptic feedback based on signals received from the Meta Quest 3 system.
2. **MetaQuest3System**
    - **Attributes**:
        - systemID: Identifier for the VR system.
        - connectedAccessories: List of accessories currently connected.
    - **Methods**:
        - integrateMotionData(): Integrates incoming motion data into VR applications.
        - sendFeedbackSignal(): Sends feedback commands to the connected accessories.
3. **User**
    - **Attributes**:
        - userID: Identifier for the individual user.
        - userPreferences: Personal settings related to accessory use and feedback strength.
    - **Methods**:
        - wearAccessory(): Attaches the accessory before starting the VR session.
        - interactWithVR(): Engages with the VR environment using data from the accessory.

3.4.4 Sequence of Interactions



3.4.5 State Diagram for Accessory Connectivity



3.5 Problem Statement

Meta Quest's motion tracking system lacks the precision needed for some more motion-heavy scenarios which has a negative effect on the user experience. An enhanced motion tracking system is necessary to improve its accuracy and reliability.

3.5.1 Feature List

Gesture Recognition - Detects rapid hand gestures with high accuracy

Real-time Calibration - Automatically adjusts tracking sensitivity by analyzing the current users' movement

Multi-Device tracking - Allows tracking across multiple devices for more accurate motion capture

Customizable Sensitivity - Allows users to adjust sensitivity settings for their specific needs

Environmental Noise Filtering - Improves motion detection by filtering unnecessary environmental information

Movement Data Storage - Keeps a log of movements in previous sessions for performance analysis

3.5.2 User Stories

As a gamer, I want precise and quick motion tracking to improve my gameplay performance

As a fitness trainer, I want to track movements so I can analyze the performance.

As a user in a crowded area, I want motion tracking to filter out irrelevant information so that only my actions are recognized

As a user with multiple devices, I want to be able to use them to improve calibration

3.5.3 Use Cases

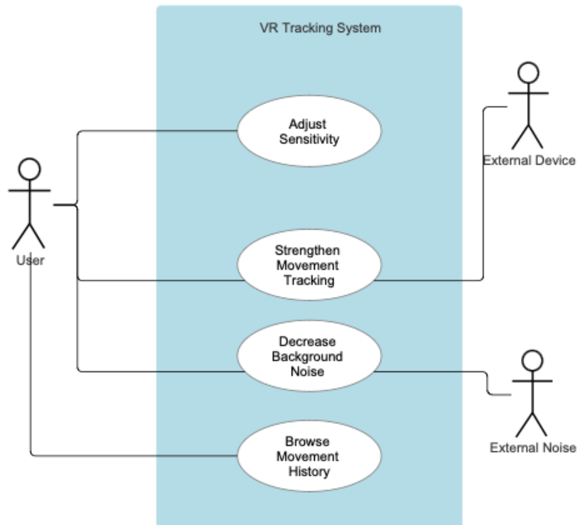
User adjusts sensitivity settings up or down (Adjust Sensitivity)

User browses the movement history from the previous session (Browse Movement History)

User uses their laptop camera to improve motion tracking (Strengthen Motion Tracking)

User puts on Meta Quest in a crowded area and enables noise filtering (Decrease Background Noise)

3.5.4 Use Case Diagram



3.6 Non Functional Requirements

1. Performance
    a. The system shall detect gestures within 25 milliseconds
    b. Motion tracking should have less than a 1-second delay

2. Reliability

    a. The system shall filter 90% of background movements/noise
    b. The system shall maintain tracking accuracy for at least 6 hours

3. Usability

    a. The user shall be able to adjust calibration while wearing VR gear
    b. Initial calibration shouldn't take more than 30 seconds
    c. Movement history should be kept as a text file

3.7 Acceptance Plan

1. Test #1 (Performance)
    - NFR: 1a
    - Test: Simulate the sequence of fast gestures and measure the time taken to recognize said gestures. The sequence should have 100 distinct gestures for accuracy

- Acceptance Criteria: The system should recognize gestures within 25 milliseconds at least 90% of the time.

2. Test #2 (Performance)
   - NFR: 1b
   - Test: Simulate a sequence of rapid movements, and measure the delay from the movement to reflection in a VR environment. Should be repeated in different environments and at different movement speeds.
   - Acceptance Criteria: The system should reflect movement in less than 1 second in 99% of cases.

3. Test #3 (Reliability)
   - NFR: 2a
   - Test: Create an environment with background movements and objects while the user performs gestures and moves around. Should have at least 5 different environments with different amounts of noise.
   - Acceptance Criteria: The system should filter at least 90% of background noise.

4. Test #4 (Reliability)
   - NFR: 2b
   - Test: Run continuous session environments with already documented user gestures every 30 minutes to track system accuracy over time.
   - Acceptance Criteria: The system should maintain the same accuracy for at least 6 hours of usage.

5. Test #6 (Usability)
   - NFR: 3a
   - Test: Have users adjust calibration while using the VR headset and track any difficulties.
   - Acceptance Criteria: Users should be able to find the calibration menu and adjust calibration without taking off gear.

6. Test #7 (Usability)
    - NFR: 3b
    - Test: Time initial calibration process in standard VR environment, repeat with different users and environments
    - Acceptance Criteria: Calibration times should be completed within 30 seconds at least 95% of the time.

7. Test #8 (Usability)
    - NFR: 3c
    - Test: Start multiple sessions and do several movements/gestures. After the session, check if the movement history file is accurate and in the correct file format.
    - Acceptance Criteria: The system should generate a .txt file containing movement data after the session is finished.

3.8 Scenarios

3.8.1 Enhanced Motion Tracking System

- Actors
    - VR Gamer, motion tracking enhancement device
- Assumptions
    - The device is compatible with Meta Quest and the software being used. The camera is currently paired with the Quest
- Happy Path
    - Gamer enables accessory and connects it to Quest
    - Gamer launches compatible VR game
    - The camera captures precise foot and hand movements and transmits them to Meta Quest
    - Meta Quest renders real-time motion in-game, improves immersion
- Error Path
    - The device fails to pair with the Quest
    - Gamer gets a notification indicating a connection error
    - The system guides the user to troubleshoot or repair the device
- Outcome
    - Gamer experiences a more immersive gaming experience with enhanced tracking

3.8.2 Browsing Movement History

- Actors
    - VR User
- Assumptions
    - Meta Quest had recorded movement data from the previous session
    - The user has access to the movement history feature in the settings
- Happy Path
    - User navigators to the movement history section in the settings
    - Meta Quest displays a log of recent movements, including details such as the time and types of certain movements
    - The user browses the movement history and selects specific sessions for a more detailed view
    - The user optionally could also export/save data as a .txt file
- Error Path
    - The user attempts to open movement history but data is either unavailable or corrupted
    - Quest displays an error message, guides the user to try again or check the settings
    - If the error continues, the system prompts users to reset their movement history
- Outcome
    - The user reviews movement history and exports data if necessary

3.8.3 Adjusting Sensitivity Settings

- Actors
    - VR User
- Assumptions
    - Meta Quest settings include adjustable sensitivity levels
    - The system supports changes without the need to remove the headset
- Happy Path
    - The user accesses VR settings menu while wearing a headset, goes to sensitivity adjustment section
    - The user selects the preferred sensitivity level
    - Quest applies the changes real-time, allowing the user to test adjustments while wearing a headset
    - The user saves settings when satisfied
- Error Path
    - User encounters delays or unresponsiveness when adjusting the sensitivity

    - Quest displays prompt suggesting troubleshooting steps such as recalibration
    - If issues continue, the system recommends restarting or resetting to default sensitivity settings
- Outcome
    - User customizes tracking sensitivity to their preference

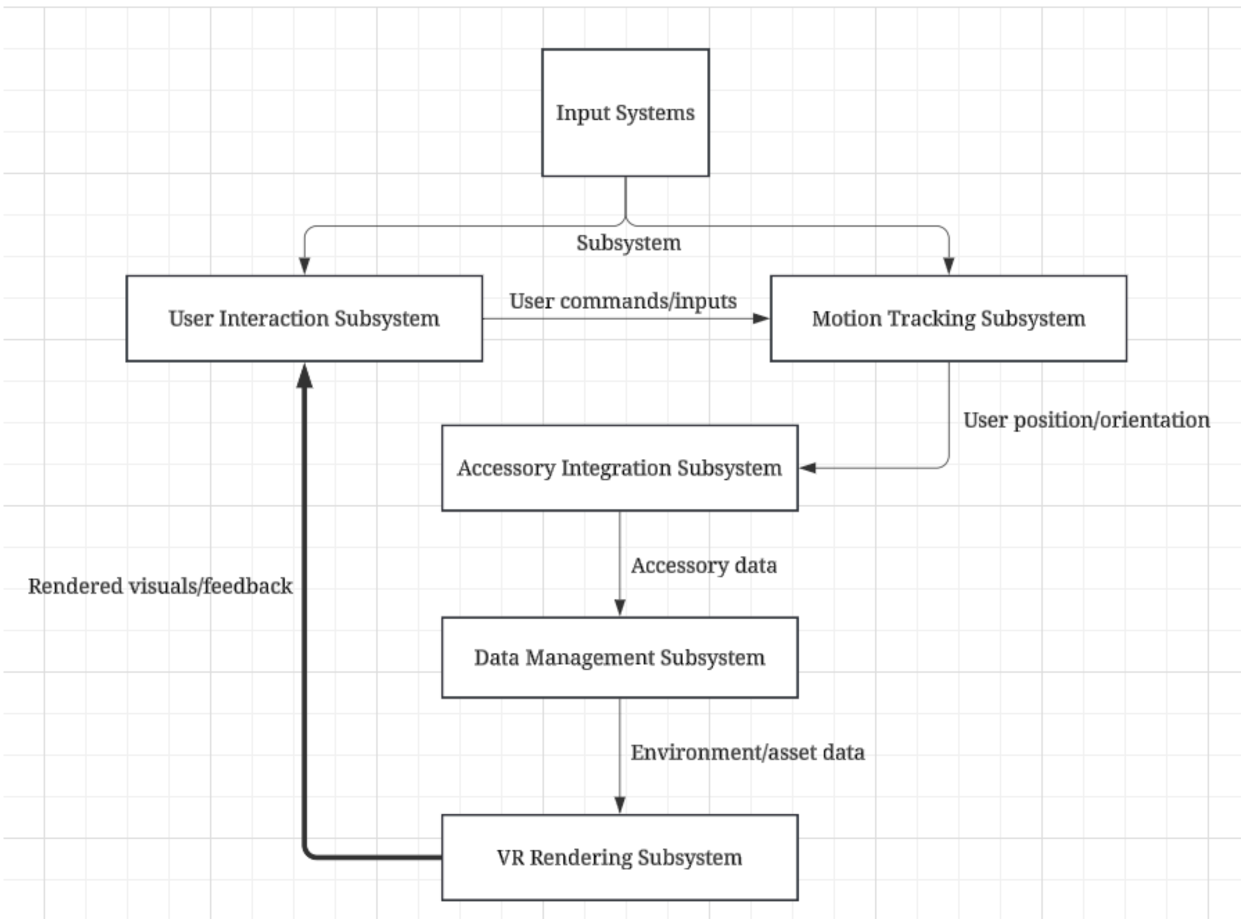
3.8.4 Decreasing Background Noise

- Actors
    - User, background people/objects
- Assumptions
    - Meta Quest has an option for reducing background noise in settings
    - The user is in an environment with substantial background motion or noise
- Happy Path
    - The user opens the VR settings menu and finds background noise filtering
    - The user enables noise filtering, which adjusts the tracking algorithm to ignore non-user movements
    - Meta Quest recalibrated tracking in real-time, reducing interference from background
    - The user continues VR activity with improved tracking accuracy due to a lack of environmental distractions
- Error Path
    - Noise filtering fails to activate or doesn't reduce background noise to a sufficient amount
    - Quest notifies the user and suggests moving to a quieter area or adjusting sensitivity manually
    - If unsuccessful, the system reverts to default tracking mode
- Outcome
    - The user enjoys enhanced tracking accuracy with reduced interference from background

4. Architecture

4.1 Subsystem Model

4.1.1 Logical Subsystems



The Meta Quest VR system can be divided into several key logical subsystems that work together to deliver an immersive and responsive virtual reality experience. Each subsystem serves a specific function and interacts with others to fulfill system requirements. Below are the key logical subsystems and their roles:

1. **User Interaction Subsystem**
   - **Role**: The User Interaction Subsystem manages all front-end interactions between the user and the VR system. It handles user authentication, and profile management, and captures user commands from controllers or other input devices.

- **Components**: This subsystem includes UI components, motion controllers, and menu navigation controls. It ensures that users can easily interact with the VR environment and that commands are translated into appropriate system actions.
- **Description**: The subsystem allows users to interact intuitively with the VR system, managing both controller-based and touch-based inputs. It connects with other subsystems to initiate actions such as calibration or environment changes.

2. **Motion Tracking Subsystem**
    - **Role**: This subsystem is responsible for accurately tracking user movement using multiple sensors embedded in the Meta Quest headset, controllers, and any external accessories.
    - **Components**: Includes sensors (gyroscopes, accelerometers, external tracking devices) and processors that capture user movement in real-time.
    - **Description**: It gathers data about the user's head, hand, and body positions and passes this information to the VR Rendering Subsystem. The goal is to provide a real-time and accurate representation of the user's actions in the virtual environment.
3. **Accessory Integration Subsystem**
    - **Role**: The Accessory Integration Subsystem connects and manages any enhanced accessories, such as motion-tracking bands or leg sensors.
    - **Components**: Bluetooth or Wi-Fi modules to establish a stable connection between the Meta Quest headset and accessories.
    - **Description**: It collects additional motion data that supplement data from the built-in sensors, providing enhanced tracking capabilities, especially for lower body movements. This subsystem aims to improve overall motion accuracy, which contributes to the immersive experience.
4. **VR Rendering Subsystem**
    - **Role**: This subsystem generates the virtual environment in response to user actions and the data received from other subsystems.
    - **Components**: Includes rendering engines, graphical processors, and display units responsible for projecting VR visuals in real time.
    - **Description**: It takes inputs from the Motion Tracking Subsystem and uses them to render corresponding changes in the virtual environment. The visuals are updated constantly to match the user's movements, ensuring a seamless VR experience.
5. **Data Management Subsystem**
    - **Role**: The Data Management Subsystem is responsible for storing user data, such as preferences, calibration settings, and movement history.

  - **Components**: Includes cloud storage, local storage units, and data management services.
  - **Description**: It ensures data persistence, allowing users to save settings for personalized experiences. It also records user movement history, which may be used for analytical purposes or improving system calibration.

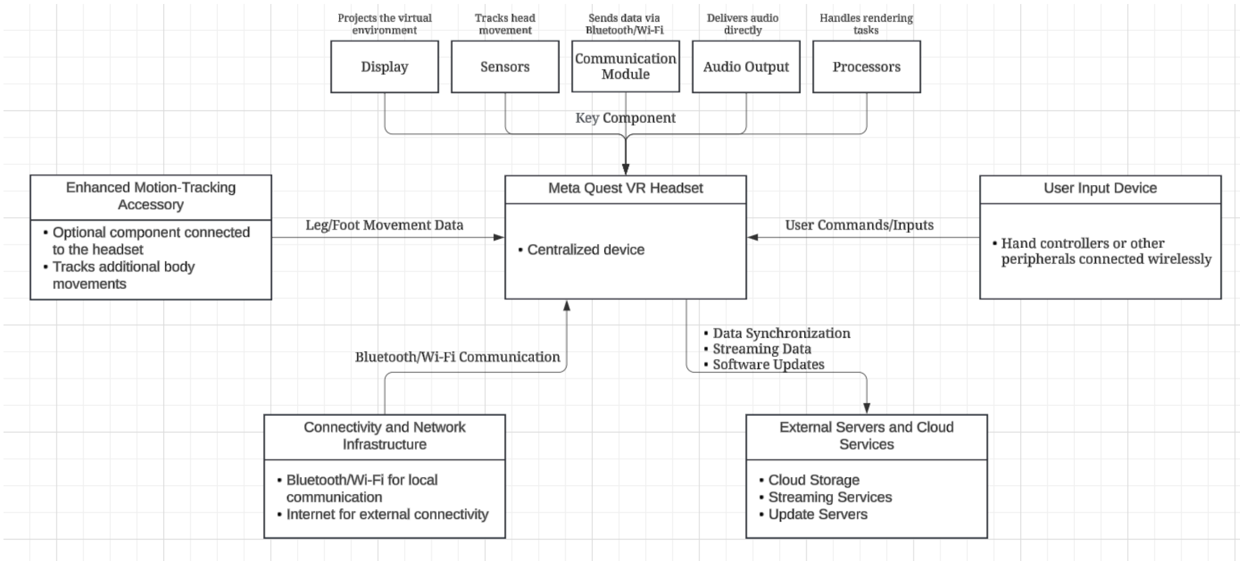
4.1.2 Subsystem Interaction

The interaction between different subsystems is critical to ensuring a responsive and immersive VR experience. Below is an overview of how each subsystem interacts with others to deliver the intended functionality.

- **User Interaction Subsystem**: Captures user input (e.g., controller actions) and sends it to the **Motion Tracking Subsystem** for real-time processing.
- **Motion Tracking Subsystem**: Gathers data from all sensors, including the **Accessory Integration Subsystem**, and passes the processed data to the **VR Rendering Subsystem**.
- **Accessory Integration Subsystem**: Collects and integrates additional movement data from external accessories, enhancing accuracy. This data is then sent to the **Motion Tracking Subsystem**.
- **VR Rendering Subsystem**: Uses data from the **Motion Tracking Subsystem** to update the visual environment displayed to the user. It directly communicates with the **Data Management Subsystem** to access user preferences.
- **Data Management Subsystem**: Stores data from other subsystems, including user preferences and calibration details, for future use. It communicates with the **User Interaction Subsystem** to retrieve stored settings and provide a personalized experience.

4.2 Target Environment

4.2.1 Target Environment Description



The target environment of the Meta Quest VR system refers to the physical and network infrastructure required to operate the virtual reality experience effectively. This includes the hardware components, communication interfaces, and external network services needed to create an immersive and seamless VR environment. Below, we provide a detailed description of the components and their interactions in the target environment.

1. **Meta Quest VR Headset**
    - **Description**: The VR headset is the core hardware component that allows users to experience the virtual environment. It includes sensors, display units, and processors required to render the VR experience.
    - **Components**:
        - **Display**: Projects the virtual environment to the user, providing a wide field of view and high resolution for immersive experiences.
        - **Sensors**: Gyroscopes, accelerometers, and external tracking cameras for detecting head movement and spatial orientation.
        - **Audio Output**: Provides spatial audio to enhance immersion.
        - **Processors**: Responsible for executing the rendering and processing tasks needed for real-time interaction.
2. **Enhanced Motion-Tracking Accessory**

- **Description**: The enhanced motion-tracking accessory is an optional add-on that provides additional sensors for improved tracking accuracy, especially for lower body movements.
- **Components**:
  - **Sensors**: Track leg and foot movement, offering additional input data to enhance the VR experience.
  - **Communication Module**: Uses Bluetooth or Wi-Fi to communicate with the VR headset in real-time, ensuring low latency.

3. **User Input Devices**
   - **Description**: These include hand controllers or other wearable devices used by the user to interact with the virtual environment.
   - **Components**:
     - **Controllers**: Provide buttons, triggers, and joysticks to allow users to navigate the VR environment and interact with virtual objects.
     - **Communication**: Connect to the headset using Bluetooth or other wireless communication methods for real-time input.
4. **Connectivity and Network Infrastructure**
   - **Description**: The communication infrastructure is crucial for connecting various devices and ensuring a seamless experience.
   - **Components**:
     - **Bluetooth/Wi-Fi Module**: Establishes communication between the VR headset, enhanced accessories, and controllers.
     - **Internet Connection**: Provides connectivity to external servers for cloud services, software updates, and data synchronization.
     - **Cloud Servers**: Store user profiles, preferences, and data, allowing the user to access their VR settings from different devices and ensure data persistence.
5. **External Servers and Cloud Services**
   - **Description**: The Meta Quest VR system relies on external servers and cloud services to store user data, stream content, and handle computationally intensive tasks.
   - **Components**:
     - **Cloud Storage**: Used for storing user preferences, movement history, and other data to enhance personalization.
     - **Streaming Services**: Provide content for VR applications, reducing the need for large local storage on the headset.
     - **Update Servers**: Deliver software updates and patches to ensure that the system stays up to date with the latest features and improvements.

4.2.3 Functional Requirements Addressed by Target Environment

- **Real-time Motion Tracking**: The target environment includes sensors, communication modules, and enhanced accessories that work together to capture user movement accurately and provide a real-time response.
- **Low Latency Communication**: The use of Bluetooth and Wi-Fi ensures low latency, allowing for seamless interaction between the user's actions and the corresponding system responses.
- **Cloud-based Personalization**: External servers store user data and preferences, allowing the system to personalize the VR experience for each user based on stored profiles.

4.2.4 Non-functional Requirements Addressed by Target Environment

Performance (Motion-to-Render Latency)

- **Requirement:** The end-to-end latency from user movement capture (sensors in headset/accessory) to updated frame rendering in the VR display must not exceed 20 milliseconds in 95% of test cases.
- **Mapped Components:** Headset processors, rendering subsystem, and wireless communication modules (Bluetooth/Wi-Fi).

Reliability (Uptime and Data Persistence)

- **Requirement:** The VR system's cloud services (including user profile storage and update servers) must achieve a monthly uptime of 99.9%, with no single outage lasting longer than 5 minutes.
- **Mapped Components:** External servers, cloud storage services, and internet connectivity.

Low-Latency Communication (User Input to Action Response)

- **Requirement:** Input from user controllers or accessories must be acknowledged and reflected in the VR environment within 10 milliseconds of signal reception in at least 95% of interactions.
- **Mapped Components:** Controllers, accessories, Bluetooth/Wi-Fi modules, and the User Interaction Subsystem.

Scalability (Concurrent User Support)

- **Requirement:** The system must support at least 10,000 concurrent cloud-service requests (e.g., retrieving stored preferences or movement history) without exceeding a 5% increase in average data retrieval latency (baseline < 50ms) during peak load conditions.
- **Mapped Components:** Cloud servers, data management subsystem, and network infrastructure.

5. Project Planning

5.1 Release Plan

Release 1: 2025-03-26

- Use Case: Developers Finish Building Prototype
    - The Prototype contains the following functions: Display, Sensors, Audio Outputs, and Processors.
    - Developers will spend a huge time testing the Enhanced Motion-Tracking Accessory.

- Use Case: Developers Finish Building Testing Model for the New Motion Tracking System
    - Developers will use the Testing Model to test the prototype.
    - The Testing Model contains simulated scenarios such as gaming, movies, and everyday use cases.

- Use Case: Alpha Testing
    - A group of developers will test the prototype including the new Motion Tracking System, Input Connectivity, Network, and Cloud server under simulated scenarios.
    - Perform other offline comprehensive testing within the internal team to identify hardware and software issues.

Release 2: 2025-06-26

- Use Case: Developers Test Motion Tracking System
    - Developers will use the Testing Model to test the prototype.
    - The Testing Model contains simulated scenarios such as gaming, movies, and everyday use cases.

- Use Case: Beta Testing for a small group of Users
    - Developers will collaborate with a small group of selected users for external testing, including the New Motion Tracking System, input connectivity, and Network.
    - Developers will join the Beta Testing and collect feedback from both developers and users.

- Use Case: Developers Test the Input Devices Connectivity
    - After the new sensors and communication module stabilize, developers will focus on testing User Input Devices.

    - Developers will test controllers, headsets, Bluetooth, or other wireless communication methods for the input.

- Use Case: Developers Refine Connectivity and Network Infrastructure
    - The communication infrastructure can connect different devices and make a synchronous experience.
    - Since various changes will be applied, the Bluetooth/Wi-Fi Module and Internet Connection Module need to be refined.

Release 3: 2025-09-26

- Use Case: Developers Refine Connectivity and Network Infrastructure
    - The communication infrastructure can connect different devices and make a synchronous experience.
    - Since various changes will be applied, the Bluetooth/Wi-Fi Module and Internet Connection Module need to be refined.

- Use Case: Official Release
    - The official release of the new product, Motion Tracking System, and other new functions.

- Use Case: Data Monitoring
    - Developers collect usage data and community feedback to improve the product experience.
    - Data science collects data from databases and servers to improve servers.

- Use Case: Data Scientists Refine External Servers and Cloud Services
    - Servers including Cloud Storage, Streaming Services, and Update Servers will be refined as the new product will soon be released and the number of users will rise.
    - There will be sufficient overhead and tooling for data scientists to begin refining and transferring data using existing data.

- Use Case: Developers Refine Connectivity and Network Infrastructure
    - The communication infrastructure can connect different devices and make a synchronous experience.
    - Since various changes will be applied, the Bluetooth/Wi-Fi Module and Internet Connection Module need to be refined.

5.2 Iteration Plan for First Release

Iteration 1

- Use Case: Developers Refine Motion Tracking System
- Goal: Refine the system based on user feedback and testing, making it more stable and efficient.
    - Set up Motion Tracking System environment
    - Set up Multiple Devices Connect environment
    - Set up Sensors and Communication MOdule environment
    - Set up Input System environment

Iteration 2

- Use Case: Data Scientists Refine External Servers and Cloud Services
- Goal: Create a system that can support three times the current number of users and ten times the current data volume. Also, it needs to connect to new module data.
    - Set up Cloud Storage for storing user preferences, movement history, and other data.
    - Connect to Streaming Services for the New Motion Tracking System.
    - Set up Update Servers for delivering software updates and patches to ensure the system stays up to date with the latest features and improvements.

Iteration 3

- Use Case: Developers Refine Connectivity and Network Infrastructure
- Goal: Standard iteration for the new product and system. Fixing existing issues, releasing patches and updates.
    - Set up a community forum for reporting issues and problems. Developers and Data Scientists could collect feedback and problems from it.
    - Developers test new patches and updates based on the Testing Model.

5.3 Risk Plan

Risk	Impact	Transition	Mitigation	Containment

		Indicator		
Negative feedback about usability or comfort	Reduced customer satisfaction /negative reviews	Reports of discomfort or usability problems during beta testing	Conduct beta testing with a diverse pool of users to get better feedback	Provide detailed user guides and customer support
Supply chain disruptions	Delayed production	Suppliers missing shipment deadlines	Diversify suppliers, secure critical components early	Scale down production to critical components while delays are happening
Competitor releasing superior VR product	Loss of market share, reduced interest in product	Competitor announcing new advancements	Emphasize Meta Quest unique selling points such as social media integration with Meta platforms	Launch promotional campaigns which show Meta Quests strengths in comparison to competitors

5.4 Project Schedule

Meta Quest Project Schedule